

Solution to Problem 2

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Abstract

This analysis theoretically investigates one-dimensional inelastic collisions among three identical beads. By applying the principles of momentum conservation and the definition of the coefficient of restitution, k , we derive explicit recurrence relations for the velocities of the beads after successive impacts. A critical coefficient of restitution, $k_c \approx 0.072$, is identified, below which an infinite number of collisions occurs. For $k > k_c$, a closed-form analytical expression for the finite total number of collisions is established. Furthermore, the limiting behaviors of the system as $k \rightarrow 0$, $k \rightarrow 1$, and $k \rightarrow k_c$ are mathematically and physically evaluated.

1 Analytical Expressions for the Velocities

Let u_n , v_n , and w_n denote the normalized velocities (scaled by v_0) of the left, middle, and right beads, respectively, immediately after the n -th impact. It is evident that these velocities must exhibit different functional forms for odd and even collisions. Bearing this in mind, the conservation of momentum and the definition of the coefficient of restitution yield the following governing equations for collisions initiated from the left:

$$u_{2n+1} + v_{2n+1} = u_{2n} + v_{2n}, \quad (1a)$$

$$u_{2n+1} - v_{2n+1} = -k(u_{2n} - v_{2n}), \quad (1b)$$

$$u_{2n+2} = u_{2n+1}, \quad (1c)$$

and analogously for collisions from the right:

$$w_{2n+2} + v_{2n+2} = w_{2n+1} + v_{2n+1}, \quad (2a)$$

$$w_{2n+2} - v_{2n+2} = -k(w_{2n+1} - v_{2n+1}), \quad (2b)$$

$$w_{2n+1} = w_{2n}. \quad (2c)$$

Since the total number of collisions depends solely on the relative velocities, it is highly advantageous to define the following relative quantities:

$$\begin{aligned} a_n &:= u_n - v_n, \\ b_n &:= w_n - v_n. \end{aligned} \quad (3)$$

Consequently, Eqs. (1b) and (2b) can be simplified to:

$$\begin{aligned} a_{2n+1} &= -ka_{2n}, \\ b_{2n+2} &= -kb_{2n+1}. \end{aligned} \quad (4)$$

Using the definitions in (3), it is also possible to combine (1a) with (1c), and (2a) with (2c), yielding:

$$\begin{aligned} 2b_{2n+1} - a_{2n+1} &= 2b_{2n} - a_{2n}, \\ 2a_{2n+2} - b_{2n+2} &= 2a_{2n+1} - b_{2n+1}. \end{aligned} \quad (5)$$

Substituting Eqs. (4) into these relations provides a more tractable set of equations:

$$\begin{aligned} a_{2n+1} &= -ka_{2n}, \\ b_{2n+1} &= -\frac{1+k}{2}a_{2n} + b_{2n}, \\ a_{2n+2} &= a_{2n+1} - \frac{1+k}{2}b_{2n+1}, \\ b_{2n+2} &= -kb_{2n+1}, \end{aligned}$$

which can be elegantly expressed in matrix form as:

$$\begin{aligned} \begin{pmatrix} a_{2n+1} \\ b_{2n+1} \end{pmatrix} &= \begin{pmatrix} -k & 0 \\ -(1+k)/2 & 1 \end{pmatrix} \begin{pmatrix} a_{2n} \\ b_{2n} \end{pmatrix}, \\ \begin{pmatrix} a_{2n+2} \\ b_{2n+2} \end{pmatrix} &= \begin{pmatrix} 1 & -(1+k)/2 \\ 0 & -k \end{pmatrix} \begin{pmatrix} a_{2n+1} \\ b_{2n+1} \end{pmatrix}. \end{aligned} \quad (6)$$

Combining these transformations results in a mapping over a full cycle of two collisions:

$$\begin{aligned} \begin{pmatrix} a_{2n+1} \\ b_{2n+1} \end{pmatrix} &= \begin{pmatrix} -k & k(1+k)/2 \\ -(1+k)/2 & (1-k)^2/4 \end{pmatrix} \begin{pmatrix} a_{2n-1} \\ b_{2n-1} \end{pmatrix}, \\ \begin{pmatrix} a_{2n+2} \\ b_{2n+2} \end{pmatrix} &= \begin{pmatrix} (1-k)^2/4 & -(1+k)/2 \\ k(1+k)/2 & -k \end{pmatrix} \begin{pmatrix} a_{2n} \\ b_{2n} \end{pmatrix}. \end{aligned} \quad (7)$$

Because the trace and determinant of both transformation matrices are identical, their eigenvalues are also identical, as anticipated from the inherent symmetry of the physical problem. The corresponding characteristic equation is given by

$$4\lambda^2 - (1 - 6k + k^2)\lambda + 4k^2 = 0, \quad (8)$$

yielding the eigenvalues:

$$\lambda_{\pm} = \frac{1 - 6k + k^2 \pm (1+k)\sqrt{1 - 14k + k^2}}{8}. \quad (9)$$

It is apparent that the argument of the square root vanishes for a critical value of the coefficient of restitution, denoted as k_c :

$$k_c = 7 - 4\sqrt{3} = \left(2 - \sqrt{3}\right)^2 \approx 0.072. \quad (10)$$

The significance of this critical value will be evaluated in the subsequent section.

Given the initial conditions $a_0 = 1$ and $b_0 = 0$, Eqs. (6) and (7) yield the following closed-form expressions after systematic algebraic manipulation:

$$a_{2n} = A_+\lambda_+^n + A_-\lambda_-^n, \quad (11a)$$

$$b_{2n} = kB_+\lambda_+^n + kB_-\lambda_-^n, \quad (11b)$$

$$a_{2n+1} = -kA_+\lambda_+^n - kA_-\lambda_-^n, \quad (11c)$$

$$b_{2n+1} = -B_+\lambda_+^{n+1} - B_-\lambda_-^{n+1}, \quad (11d)$$

Table 1 Analytical expressions for the normalized velocities of the beads during the first few collisions.

n	0	1	2	3	4
a_n	1	$-k$	$\frac{1-2k+k^2}{4}$	$\frac{-k+2k^2-k^3}{4}$	$\frac{1-8k-2k^2-8k^3+k^4}{16}$
b_n	0	$\frac{-1-k}{2}$	$\frac{k+k^2}{2}$	$\frac{-1+5k+5k^2-k^3}{8}$	$\frac{k-5k^2-5k^3+k^4}{8}$
u_n	1	$\frac{1-k}{2}$	$\frac{1-k}{2}$	$\frac{3-3k+k^2-k^3}{8}$	$\frac{3-3k+k^2-k^3}{8}$
v_n	0	$\frac{1+k}{2}$	$\frac{1-k^2}{4}$	$\frac{3-k-3k^2+k^3}{8}$	$\frac{5+2k+4k^2+6k^3-k^4}{16}$
w_n	0	0	$\frac{1+2k+k^2}{4}$	$\frac{1+2k+k^2}{4}$	$\frac{5+4k-6k^2-4k^3+k^4}{16}$

where the coefficients are defined as:

$$A_{\pm} \equiv \frac{1}{2} \pm \frac{1+k}{2\sqrt{1-14k+k^2}}, \quad (12a)$$

$$B_{\pm} \equiv \pm \frac{2}{\sqrt{1-14k+k^2}}. \quad (12b)$$

The first few velocities derived from these formulations are presented in Tab. 1.

2 Number of Collisions

According to the definitions in (3), the collision sequence naturally terminates when either $a_{2n} \leq 0$ or $b_{2n+1} \geq 0$; let us denote this terminal state by $n = n^*$. In the first scenario, where the final collision occurs from the right, the total number of collisions is $N = 2n^*$. Conversely, in the second scenario, where the final collision occurs from the left, $N = 2n^* + 1$.¹ Therefore, based on Eqs. (11a) and (11d), the sequence terminates when the logical condition $\mathcal{S} = \mathcal{S}_1 \vee \mathcal{S}_2$ is met, where:

$$\mathcal{S}_1 \equiv \exists n \in \mathbb{N} : A_+ \lambda_+^n + A_- \lambda_-^n \leq 0, \quad (13a)$$

$$\mathcal{S}_2 \equiv \exists n \in \mathbb{N} : B_+ \lambda_+^{n+1} + B_- \lambda_-^{n+1} \leq 0. \quad (13b)$$

We shall now demonstrate that the total number of collisions is fundamentally governed by whether $k \leq k_c$ or $k > k_c$, with k_c given by (10). In the former regime, the sequence is infinite, and the beads asymptotically approach a common normalized velocity of $1/3$. In the latter regime, the number of collisions is strictly finite.

First, consider the regime where $k \leq k_c$. Here, both eigenvalues are strictly real. From Eq. (8), it is clear that $\lambda_+ + \lambda_- = (1 - 6k + k^2)/4 > 0$ for $k \leq k_c$ and $\lambda_+ \lambda_- = k^2 \geq 0$. Consequently, $\lambda_+ > \lambda_- \geq 0$. Notice from Eq. (12a) that $A_+ > 0$ and $A_+ + A_- = 1$. To assess the validity of (13a), we can express the condition as:

$$A_+ (\lambda_+^n - \lambda_-^n) + \lambda_-^n > 0,$$

¹It should be noted that we seek to minimize N , rather than merely n . Formulating the problem in terms of N will prove mathematically convenient in subsequent steps.

which remains strictly positive for all $n \in \mathbb{N}$. Therefore, $\mathcal{S}_1$ can never be satisfied. An analogous argument holds for $\mathcal{S}_2$: from (12b), $B_+ = -B_- > 0$, and since $\lambda_+ > \lambda_-$, $\mathcal{S}_2$ is likewise never satisfied. As a result, for $k \leq k_c$, the total number of collisions is infinite.

Next, we determine the finite number of collisions for the regime $k > k_c$. Under these conditions, the eigenvalues form a complex conjugate pair:

$$\lambda_{\pm} = \frac{1 - 6k + k^2 \pm i(1+k)\sqrt{-1+14k-k^2}}{8}.$$

It readily follows that $|\lambda_+| = |\lambda_-| = k$,² and $\angle\lambda_+ = -\angle\lambda_-$. We may thus write:

$$\lambda_{\pm} = ke^{\pm i\alpha}, \quad \text{where} \quad \alpha \equiv \tan^{-1} \left(\frac{(1+k)\sqrt{-1+14k-k^2}}{1-6k+k^2} \right). \quad (14)$$

Here, the principal *positive* branch of the arctangent is assumed.

Applying a similar polar representation to $A_{\pm}$ in (12a), noting that $|A_+| = |A_-|$, we obtain:

$$A_{\pm} = |A_{\pm}|e^{\pm i\beta}, \quad \text{where} \quad \beta \equiv -\tan^{-1} \left(\frac{1+k}{\sqrt{-1+14k-k^2}} \right). \quad (15)$$

Furthermore, $|B_+| = |B_-|$ and $\angle B_{\pm} = \mp\pi/2$. Substituting these polar forms into conditions (13), the objective reduces to finding the minimal N satisfying:

$$\cos(n\alpha + \beta) \leq 0 \vee \sin(n\alpha + \alpha) \leq 0.$$

These inequalities demand $n\alpha + \beta \geq \pi/2$ and $n\alpha + \alpha \geq \pi$, respectively. Consequently, mapping back to $N = 2n$ and $N = 2n + 1$ for the respective conditions yields:

$$N(k > k_c) = \min \left\{ \left\lceil \frac{\pi - 2\beta}{\alpha} \right\rceil, \left\lceil \frac{2\pi - \alpha}{\alpha} \right\rceil \right\}.$$

Although not immediately obvious, one can demonstrate using Eqs. (14) and (15) that $\alpha - 2\beta = \pi$. Utilizing this identity, the two arguments inside the minimum function are precisely identical. Hence, we arrive at the elegant analytical result:

$$N(k > k_c) = \left\lceil 2\pi \tan^{-1} \left(\frac{(1+k)\sqrt{-1+14k-k^2}}{1-6k+k^2} \right)^{-1} \right\rceil - 1. \quad (16)$$

This discrete function is illustrated in Fig. 1.

3 Limiting Cases $k \rightarrow 0$, $k \rightarrow 1$, and $k \rightarrow k_c$

In this section, we rigorously examine the limiting cases of the system to verify their alignment with physical intuition.

First, consider the limit as $k \rightarrow 0$. As proven in the previous section, $N(k \leq k_c) = \infty$, and the absolute inelastic limit $k \rightarrow 0$ is no exception. At

²This magnitude equality is a direct consequence of the last term in (8).

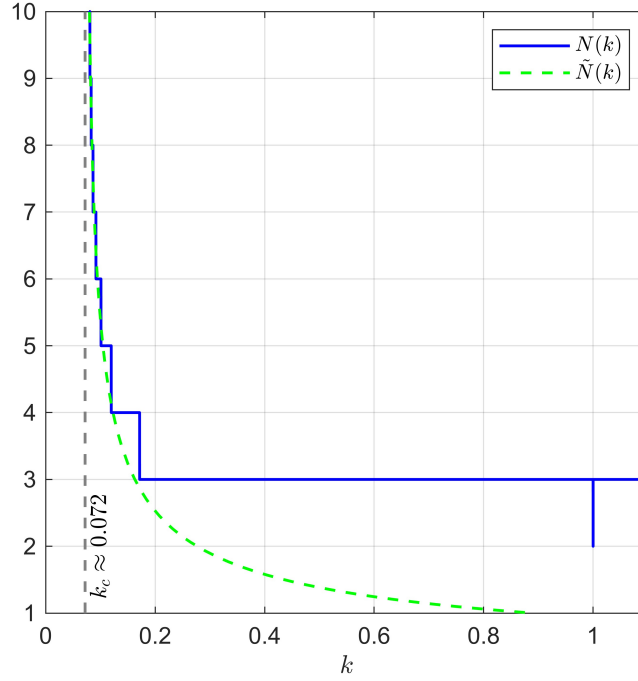


Figure 1 Total number of collisions, N , as a function of the coefficient of restitution, k .

first glance, this might appear to contradict the standard assumption that in perfectly inelastic collisions, bodies must instantly stick together. However, this is mathematically imprecise; the definition of the coefficient of restitution for perfectly inelastic collisions only dictates that the relative velocity of the bodies after impact vanishes. It does not inherently enforce spatial merging (i.e., they might maintain a minute separation). Because our foundational equations in Section 1 assumed that each bead participates in the collision sequence as an independent entity, recovering an infinite collision count in this mathematical limit is entirely consistent.

Next, we evaluate the system as $k \rightarrow 1$. Let us define $k = 1 + \epsilon$ with $|\epsilon| \ll 1$.³ Expanding (16) under this assumption yields:

$$N(k = 1 + \epsilon) \approx 2 + \left\lceil \frac{3\sqrt{3}}{8\pi} \epsilon^2 \right\rceil.$$

This indicates that the limits from both sides agree:

$$\lim_{k \rightarrow 1^+} N(k) = \lim_{k \rightarrow 1^-} N(k) = 3,$$

whereas the exact evaluation yields $N(k = 1) = 2$. Thus, the function $N(k)$ exhibits a mathematical discontinuity at $k = 1$. Physically, this occurs because when k is marginally close to unity, the leftmost bead retains a minute rightward velocity after interaction (see Tab. 1), eventually causing a third collision.

³Analyzing coefficients of restitution greater than unity is theoretically permissible and corresponds physically to energy-releasing explosions occurring upon impact.

Only in the perfectly elastic case ($k = 1$) does the left bead's velocity become identically zero after the primary impact.

Finally, we analyze the transition near the critical point by setting $k = k_c + \epsilon$, where ϵ is a small positive parameter. A local expansion of (16) in the vicinity of k_c provides the asymptotic behavior:

$$N(k = k_c + \epsilon) \sim \tilde{N}(k) \equiv \pi \sqrt{\frac{14\sqrt{3} - 24}{3\epsilon}} \approx 0.90\epsilon^{-1/2}.$$

This singular asymptotic divergence is vividly depicted in Fig. 1.